Questions and marks

Candidates will select and answer one physical question from Section A and one human question from Section B based on their chosen topics. The questions require longer responses, each with three parts, designed to include data response, investigation and evaluative skills and related impacts/management issues. Each question will be worth 35 marks.

Candidates will be expected to use the resource booklet provided, their own ideas, relevant fieldwork and research which they have carried out. Candidates must not take materials into the examination.

There are a total of 70 marks for the examination.

2.3 Topic 1: Extreme Weather

Extreme weather

Extreme weather includes a range of phenomena that involve extremes of temperature, precipitation, wind and atmospheric pressure. They in turn develop from a variety of meteorological conditions. This topic looks at how extreme weather events lead to immediate, subsequent and longer term hazards. Storms, river floods and drought clearly illustrate the environmental, social and economic impacts of extreme weather — impacts that are closely related to the type of hazard involved and the economic situation of those affected. Risks from extreme weather, such as flooding, are increasing and much of this is our fault. If extreme weather conditions are becoming more frequent and more severe, then tougher, fairer and more intelligent decisions will need to be taken in both the short and longer term.

Fieldwork and research opportunities

Fieldwork opportunities include a weather log, flood impacts survey, flood/drought risk assessments and flood management assessments**. Research work** could relate to weather records, satellite images, hurricane data, and use of statistics for flood/drought events as well as evaluations of various management strategies.

1 Extreme weather watch

Enquiry question: *What are extreme weather conditions and how and why do they lead to extreme weather events?*

What students need to learn	Suggested teaching and learning
■ There is a wide variety of extreme (severe or unexpected) weather phenomena.	■ Defining and examining the nature and distribution of different types of extreme weather, such as tropical cyclones, temperate storms, tornadoes, flooding, blizzards, winter weather, heat waves, fires, and drought.
■ **Fieldwork and research**, using a weather diary and synoptic maps, into meteorological conditions (air masses, pressure systems and fronts) which can influence changes in temperatures, precipitation and winds. These lead to contrasting weather events such as the development of a depression or seasonal anticyclones.	■ Using primary and secondary sources to monitor and understand how differing weather patterns relate to underlying meteorological conditions.
■ Contrasting examples of how extreme weather conditions develop such as hurricanes, snow and ice, and drought.	■ Researching meteorological processes such as a hurricane sequence, UK or USA winter conditions and an extended drought.

2 Extreme impacts

Enquiry question: *What are the impacts of extreme weather on people, the economy and the environment?*

What students need to learn	Suggested teaching and learning
■ An extreme weather hazard can have different impacts depending on the severity of the event, a location's level of economic development and the vulnerability of those affected.	■ Researching how the impacts of extreme weather vary in intensity and in different parts of the world.

■ **Fieldwork and research** into the social, economic and environmental impacts of extreme weather created by: ◆ an immediate disastrous weather event — such as a tornado or hurricane ◆ a subsequent additional hazard — such as localised river flooding ◆ a longer-term trend or condition — such as a heat wave or drought.	■ Using primary and secondary sources to investigate impacts on homes, business, health, lives, infrastructure, production and habitats. ■ Examining specific examples of the impacts of extreme weather events, such as Hurricane Mitch, a localised flood event and drought in New South Wales as well as examples relating to similar current events.

3 Increasing risks

Enquiry question: *How are people and places increasingly at risk from and vulnerable to extreme weather?*

What students need to learn	Suggested teaching and learning
■ Evidence that extreme weather hazards in the UK and elsewhere are becoming more frequent and involve higher risk due to natural and human causes such as climate change, demographics and land management.	■ Investigating how the increased incidence and risk of weather hazards is affected by climate change, global warming, population growth along rivers and coastlines, and poor management of land.
■ **Fieldwork and research** to investigate how a small stream or part of a river catchment can suffer increased flood risks resulting from: ◆ meteorological causes ◆ the physical characteristics of the area ◆ growing urbanisation, land use change and attempts at management.	■ Using primary and secondary sources to investigate and analyse a range of causes of increased flood risks at a local scale such as in Carlisle or Uckfield, including for example: ◆ heavy/prolonged precipitation or snow melt ◆ geology, vegetation and slopes ◆ land use and management.

4 Managing extreme weather

Enquiry question: *How can we best respond to and cope with the impacts of extreme weather?*

What students need to learn	Suggested teaching and learning
■ **Fieldwork and research** into ways of managing and responding to extreme weather events using short- and longer-term strategies, and how some management strategies are more successful than others.	■ Using primary and secondary sources to investigate strategies such as USA hurricane warning, Environment Agency flood protection and risk assessments at a local scale, eg York.
■ The role of new technology in improving community preparedness, event forecasting and reducing impacts of disasters.	■ Researching the role of technology and its application to extreme weather management such as forecasting (NOAA), flood monitoring or the use of drought resistant crops.
■ Ways to manage drought through physical, social, economic and political responses in contrasting areas.	■ Assessing sustainable longer-term solutions for tackling drought such as water management and adapting farming techniques as in south east England or Ethiopia.